🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.7

Generate Equivalent Expressions

Lesson 6-8 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

TODAY'S OBJECTIVES

Content Objective: I can use the commutative, associative, and identity properties to rewrite expressions.
Language Objective: I can explain my reasoning using the words commutative property, associative property, and identity property.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Properties of Operations | Commutative Property | Associative Property | Identity Property | Property | Distributive Property

Tap any word to see what it means and a picture.

1. Rules that let you reorder or regroup without changing the ________.

2. The rule that order does not change a sum or product, like a + b = b + a, is the ________.

3. The rule that grouping does not change a sum or product, like (a + b) + c = a + (b + c), is the ________.

4. The rule that adding 0 or multiplying by 1 keeps a number the same is the ________.

5. A rule that is always true in math is a ________.

6 The rule that lets you multiply a number across a sum, like $a(b + c) = a \times b + a \times c$, is the ________.

Write About the Math **Guided ESOL writing**

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What properties did the engineer use, and why does rearranging make the calculation easier?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Properties of Operations** — Rules that let you reorder or regroup numbers without changing an expression's value.
  Propiedades de las operaciones — Reglas que permiten reordenar o reagrupar números sin cambiar el valor de una expresión.

- [ ] **Commutative Property** — You can change the order and get the same answer.
  Propiedad conmutativa — Puedes cambiar el orden y obtener la misma respuesta.

- [ ] **Associative Property** — You can change the grouping and get the same answer.
  Propiedad asociativa — Puedes cambiar la agrupación y obtener la misma respuesta.

- [ ] **Identity Property** — Adding 0 or multiplying by 1 keeps the same value.
  Propiedad de identidad — Sumar 0 o multiplicar por 1 mantiene el mismo valor.

- [ ] **Property** — A rule that is always true in math.
  Propiedad — Una regla que siempre es verdadera en matemáticas.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The engineer used the ____ property to change the ____ and the ____ property to change the ____. This makes it easier because $4 \times 25 =$ ____, which is a friendly number to multiply by.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The commutative property lets you change the order of a sum without changing the total.

Evidence: Students name the commutative property and explain that 3 + 5 + 2 gives 10 in any order because reordering addends does not change the sum.

Reasoning: This evidence connects the definition of properties of operations to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The engineer used the ____ property to change the ____ and the ____ property to change the ____. This makes it easier because $4 \times 25 =$ ____, which is a friendly number to multiply by.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Properties of Operations
2. **Notes 2:** Commutative Property
3. **Notes 3:** Associative Property
4. **Notes 4:** Identity Property
5. **Notes 5:** Property
6. **Notes 6:** Distributive Property
7. **Watch for this mistake:** *A common mistake in Properties of Operations is applying the commutative property where it does not belong, or confusing it with the associative property. Students often assume 15 − 8 = 8 − 15 (it does not: 7 ≠ −7 — commutative only works for addition and multiplication), or they see parentheses move and call it "associative" even when the numbers inside just swapped places, like (2 + 3) + 4 = (3 + 2) + 4 (that is commutative — the order inside the parentheses changed, not the grouping). Before you label a property, ask: did only the order change (commutative), did only the grouping change (associative), or was a 0 or 1 involved (identity)?*
8. **Try It 1:** A. Associative Property — *The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.*
9. **Try It 2:** A. Identity Property — *Multiplying by 1 keeps the same value, so this is the Identity Property of Multiplication.*